

GARP Risk & AI — 100 Simple Numerical Problems

Unofficial Exam Prep — With Detailed Definitions

Unofficial

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Question

A model has $TP = 40$, $TN = 50$, $FP = 10$, $FN = 20$. What is the accuracy?

- A. 0.50
- B. 0.75
- C. 0.80
- D. 0.90

Answer: B

Explanation

Definition: Accuracy is the proportion of all predictions (both positive and negative) that the model classified correctly. It is the most intuitive performance measure but can be misleading when classes are imbalanced.

Formula: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$

Calculation: $(40 + 50) / (40 + 50 + 10 + 20) = 90 / 120 = 0.75$

Interpretation: The model correctly classified 75 percent of all instances.

Question

A model has $TP = 30$, $FP = 20$. What is the precision?

- A. 0.40
- B. 0.50
- C. 0.60
- D. 0.75

Answer: C

Explanation

Definition: Precision (also called Positive Predictive Value) is the proportion of positive predictions that are actually correct. It answers the question: "Of all the cases the model flagged as positive, how many were truly positive?" High precision means few false alarms.

Formula: $\text{Precision} = TP / (TP + FP)$

Calculation: $30 / (30 + 20) = 30 / 50 = 0.60$

Interpretation: When the model predicts positive, it is correct 60 percent of the time.

Question

A model has $TP = 25$, $FN = 75$. What is the recall?

- A. 0.25
- B. 0.33
- C. 0.50
- D. 0.75

Answer: A

Explanation

Definition: Recall (also called Sensitivity or True Positive Rate) is the proportion of actual positive cases that the model correctly identified. It answers: "Of all the truly positive cases, how many did the model catch?" High recall means few missed positives.

Formula: $\text{Recall} = TP / (TP + FN)$

Calculation: $25 / (25 + 75) = 25 / 100 = 0.25$

Interpretation: The model only caught 25 percent of all actual positive cases; it missed 75 percent.

Question

A model has precision = 0.60 and recall = 0.40. What is the F1 score?

- A. 0.24
- B. 0.48
- C. 0.50
- D. 0.60

Answer: B

Explanation

Definition: The F1 score is the harmonic mean of precision and recall. It combines both metrics into a single value, rewarding models that balance precision and recall. It is especially useful when classes are imbalanced. The harmonic mean penalizes extreme values (e.g., very high precision but very low recall).

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Calculation: $2 \times (0.60 \times 0.40) / (0.60 + 0.40) = 2 \times 0.24 / 1.00 = 0.48$

Interpretation: The F1 score of 0.48 reflects a moderate balance between precision and recall.

Question

A model has accuracy = 0.82. What is the error rate?

- A. 0.08
- B. 0.18
- C. 0.22
- D. 0.28

Answer: B

Explanation

Definition: The error rate (also called misclassification rate) is the proportion of all predictions that the model got wrong. It is the complement of accuracy.

Formula: $\text{Error Rate} = 1 - \text{Accuracy} = (\text{FP} + \text{FN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

Calculation: $1 - 0.82 = 0.18$

Interpretation: 18 percent of all predictions were incorrect.

Question

A model has $TN = 80$, $FP = 20$. What is the specificity?

- A. 0.20
- B. 0.50
- C. 0.80
- D. 0.90

Answer: C

Explanation

Definition: Specificity (also called True Negative Rate) is the proportion of actual negative cases that the model correctly identified as negative. It answers: "Of all the truly negative cases, how many did the model correctly leave alone?" High specificity means few false alarms on the negative class.

Formula: $\text{Specificity} = TN / (TN + FP)$

Calculation: $80 / (80 + 20) = 80 / 100 = 0.80$

Interpretation: The model correctly identified 80 percent of all actual negative cases.

Question

A model has $TP = 15$, $TN = 70$, $FP = 5$, $FN = 10$. What is the accuracy?

- A. 0.75
- B. 0.80
- C. 0.85
- D. 0.90

Answer: C

Explanation

Definition: Accuracy measures the overall proportion of correct predictions (both positive and negative).

Formula: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$

Calculation: $(15 + 70) / (15 + 70 + 5 + 10) = 85 / 100 = 0.85$

Interpretation: 85 percent of all predictions were correct.

Question

A model has $TP = 45$, $FP = 15$. What is the precision?

- A. 0.60
- B. 0.70
- C. 0.75
- D. 0.80

Answer: C

Explanation

Definition: Precision is the proportion of positive predictions that are correct.

Formula: $\text{Precision} = TP / (TP + FP)$

Calculation: $45 / (45 + 15) = 45 / 60 = 0.75$

Interpretation: When the model predicts positive, it is correct 75 percent of the time.

Question

A model has $TP = 60$, $FN = 40$. What is the recall?

- A. 0.40
- B. 0.50
- C. 0.60
- D. 0.75

Answer: C

Explanation

Definition: Recall is the proportion of actual positive cases that the model correctly identified.

Formula: $\text{Recall} = TP / (TP + FN)$

Calculation: $60 / (60 + 40) = 60 / 100 = 0.60$

Interpretation: The model caught 60 percent of all actual positive cases.

Question

A model has precision = 0.80 and recall = 0.50. What is the F1 score?

- A. 0.40
- B. 0.50
- C. 0.62
- D. 0.80

Answer: C

Explanation

Definition: The F1 score is the harmonic mean of precision and recall, providing a single balanced metric.

Formula: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Calculation: $2 \times (0.80 \times 0.50) / (0.80 + 0.50) = 2 \times 0.40 / 1.30 = 0.80 / 1.30 = 0.615 \approx 0.62$

Interpretation: The F1 score of 0.62 reflects a moderate balance, penalized by the lower recall.

Question

Actual values: [10, 20, 30]. Predicted values: [12, 18, 33]. What is the MSE?

- A. 4.33
- B. 5.67
- C. 7.33
- D. 8.67

Answer: B

Explanation

Definition: Mean Squared Error (MSE) is the average of the squared differences between actual and predicted values. Squaring ensures positive and negative errors do not cancel out and heavily penalizes large errors.

Formula: $MSE = (1/N) \times \sum (y_i - \hat{y}_i)^2$

Calculation: Errors: $(10-12)=-2$, $(20-18)=2$, $(30-33)=-3$. Squared: 4, 4, 9. Sum = 17. $MSE = 17/3 = 5.67$

Interpretation: On average, the squared prediction error is 5.67.

Question

MSE = 25. What is the RMSE?

- A. 3
- B. 4
- C. 5
- D. 6

Answer: C

Explanation

Definition: Root Mean Squared Error (RMSE) is the square root of MSE. It is expressed in the same units as the target variable, making it more interpretable than MSE.

Formula: $RMSE = \sqrt{MSE}$

Calculation: $\sqrt{25} = 5$

Interpretation: The typical prediction error is about 5 units (in the same units as the target).

Question

Actual values: [5, 10, 15]. Predicted values: [7, 8, 18]. What is the MAE?

- A. 2.00
- B. 2.33
- C. 2.67
- D. 3.00

Answer: B

Explanation

Definition: Mean Absolute Error (MAE) is the average of the absolute differences between actual and predicted values. Unlike MSE, it does not square errors, so it is more robust to outliers.

Formula: $MAE = (1/N) \times \sum |y_i - \hat{y}_i|$

Calculation: $|5-7|=2$, $|10-8|=2$, $|15-18|=3$. Sum = 7. $MAE = 7/3 = 2.33$

Interpretation: On average, predictions are off by 2.33 units.

Question

Actual = 100, Predicted = 110. What is the absolute percentage error for this observation?

- A. 5 percent
- B. 10 percent
- C. 11 percent
- D. 20 percent

Answer: B

Explanation

Definition: Absolute Percentage Error (APE) measures the error as a percentage of the actual value. It is scale-independent, allowing comparison across different magnitudes. MAPE is the average of APE across all observations.

Formula: $APE = |y - \hat{y}| / y \times 100 \text{ percent}$

Calculation: $|100 - 110| / 100 = 10/100 = 0.10 = 10 \text{ percent}$

Interpretation: The prediction is 10 percent higher than the actual value.

Question

Errors are: 2, -4, 6. What is the MSE?

- A. 4.67
- B. 6.67
- C. 8.67
- D. 18.67

Answer: D

Explanation

Definition: MSE is the average of squared errors. Given errors directly, we square each and average.

Formula: $MSE = (1/N) \times \sum (\text{error}_i)^2$

Calculation: Squared errors: 4, 16, 36. Sum = 56. $MSE = 56/3 = 18.67$

Interpretation: The average squared error is 18.67.

Question

Errors are: 1, 3, 2. What is the RMSE?

- A. 1.67
- B. 2.16
- C. 3.00
- D. 4.67

Answer: B

Explanation

Definition: RMSE is the square root of MSE, expressed in original units.

Formula: $RMSE = \sqrt{(1/N) \times \sum(\text{error}_i)^2}$

Calculation: Squared errors: 1, 9, 4. Sum = 14. $MSE = 14/3 = 4.67$. $RMSE = \sqrt{4.67} = 2.16$

Interpretation: The typical error magnitude is 2.16 units.

Question

Errors are: -3, 5, -2, 4. What is the MAE?

- A. 2.50
- B. 3.00
- C. 3.50
- D. 4.00

Answer: C

Explanation

Definition: MAE is the average of absolute errors, treating positive and negative errors equally.

Formula: $MAE = (1/N) \times \sum |error_i|$

Calculation: Absolute errors: 3, 5, 2, 4. Sum = 14. $MAE = 14/4 = 3.50$

Interpretation: On average, predictions are off by 3.50 units.

Question

Actual values: [50, 100]. Predicted values: [55, 90]. What is the MAPE?

- A. 10 percent
- B. 12 percent
- C. 15 percent
- D. 20 percent

Answer: A

Explanation

Definition: Mean Absolute Percentage Error (MAPE) is the average of absolute percentage errors. It expresses forecast error as a percentage of actual values.

Formula: $MAPE = (1/N) \times \sum |(y_i - \hat{y}_i)/y_i| \times 100 \text{ percent}$

Calculation: $APE1 = |50 - 55|/50 = 5/50 = 10 \text{ percent}$. $APE2 = |100 - 90|/100 = 10/100 = 10 \text{ percent}$. $MAPE = (10 + 10)/2 = 10 \text{ percent}$

Interpretation: On average, predictions deviate from actuals by 10 percent.

Question

Actual values: [8, 12]. Predicted values: [10, 10]. What is the MSE?

- A. 2
- B. 4
- C. 8
- D. 16

Answer: B

Explanation

Definition: MSE is the average of squared errors.

Formula: $MSE = (1/N) \times \sum (y_i - \hat{y}_i)^2$

Calculation: Errors: $8-10=-2$, $12-10=2$. Squared: 4, 4. Sum = 8. $MSE = 8/2 = 4$

Interpretation: The average squared prediction error is 4.

Question

MSE = 16. What is the RMSE?

- A. 2
- B. 4
- C. 8
- D. 16

Answer: B

Explanation

Definition: RMSE is the square root of MSE, expressed in original units.

Formula: $RMSE = \sqrt{MSE}$

Calculation: $\sqrt{16} = 4$

Interpretation: The typical prediction error is 4 units.

Question

$P(A) = 0.3$, $P(B|A) = 0.8$, $P(B) = 0.4$. What is $P(A|B)$?

- A. 0.24
- B. 0.40
- C. 0.60
- D. 0.80

Answer: C

Explanation

Definition: Bayes' theorem updates the probability of an event A given new evidence B. It combines the prior probability $P(A)$, the likelihood $P(B|A)$, and the evidence $P(B)$.

Formula: $P(A|B) = P(B|A) \times P(A) / P(B)$

Calculation: $0.8 \times 0.3 / 0.4 = 0.24 / 0.4 = 0.60$

Interpretation: Given that B occurred, there is a 60 percent probability that A also occurred.

Question

$P(A) = 0.5$, $P(B|A) = 0.6$, $P(B) = 0.3$. What is $P(A|B)$?

- A. 0.50
- B. 0.60
- C. 0.80
- D. 1.00

Answer: D

Explanation

Definition: Bayes' theorem updates prior beliefs with new evidence.

Formula: $P(A|B) = P(B|A) \times P(A) / P(B)$

Calculation: $0.6 \times 0.5 / 0.3 = 0.30 / 0.3 = 1.00$

Interpretation: Given B, A is certain to occur (in this simplified example).

Question

Prior $P(\text{Good}) = 0.6$. Likelihoods: $P(w_1|\text{Good}) = 0.5$, $P(w_2|\text{Good}) = 0.4$. What is the unnormalized posterior?

- A. 0.06
- B. 0.12
- C. 0.24
- D. 0.40

Answer: B

Explanation

Definition: Naive Bayes calculates the probability of a class given observed features, assuming features are conditionally independent. The unnormalized posterior is the product of the prior and all likelihoods.

Formula: $\text{Unnormalized Posterior} = P(\text{Class}) \times \prod P(\text{word}_i|\text{Class})$

Calculation: $0.5 \times 0.4 \times 0.6 = 0.12$

Interpretation: The unnormalized score for "Good" is 0.12 (to be normalized against other classes).

Question

Prior $P(\text{Bad}) = 0.4$. Likelihoods: $P(w_1|\text{Bad}) = 0.7$, $P(w_2|\text{Bad}) = 0.3$. What is the unnormalized posterior?

- A. 0.084
- B. 0.120
- C. 0.210
- D. 0.280

Answer: A

Explanation

Definition: The unnormalized posterior for a class is the product of its prior and the likelihoods of observed features.

Formula: $\text{Unnormalized Posterior} = P(\text{Class}) \times \prod P(\text{word}_i|\text{Class})$

Calculation: $0.7 \times 0.3 \times 0.4 = 0.084$

Interpretation: The unnormalized score for "Bad" is 0.084.

Question

Unnormalized posteriors: Good = 0.12, Bad = 0.08. What is the normalized P(Good)?

- A. 0.40
- B. 0.50
- C. 0.60
- D. 0.67

Answer: C

Explanation

Definition: Normalization converts unnormalized posterior scores into proper probabilities that sum to 1.

Formula: $P(\text{Good}) = \text{Unnorm}(\text{Good}) / (\text{Unnorm}(\text{Good}) + \text{Unnorm}(\text{Bad}))$

Calculation: $0.12 / (0.12 + 0.08) = 0.12 / 0.20 = 0.60$

Interpretation: Given the observed words, there is a 60 percent probability the document belongs to the "Good" class.

Question

Data: [4, 8, 12, 16]. What is the mean?

- A. 8
- B. 10
- C. 12
- D. 14

Answer: B

Explanation

Definition: The mean (average) is the sum of all values divided by the number of values. It measures the central tendency of the data.

Formula: $\text{Mean} = (\sum x_i) / N$

Calculation: $(4+8+12+16) / 4 = 40 / 4 = 10$

Interpretation: The average value in the dataset is 10.

Question

Data: [2, 4, 6]. What is the population variance?

- A. 1.33
- B. 2.67
- C. 4.00
- D. 8.00

Answer: B

Explanation

Definition: Variance measures the average squared deviation from the mean. It quantifies how spread out the data is.

Formula: $\text{Variance} = (1/N) \times \sum (x_i - \mu)^2$

Calculation: Mean = 4. Squared deviations: $(2-4)^2=4$, $(4-4)^2=0$, $(6-4)^2=4$. Sum = 8. Variance = $8/3 = 2.67$

Interpretation: The data points deviate from the mean by an average squared amount of 2.67.

Question

Variance = 9. What is the standard deviation?

- A. 2
- B. 3
- C. 4
- D. 9

Answer: B

Explanation

Definition: Standard deviation is the square root of variance. It is expressed in the same units as the data, making it easier to interpret.

Formula: $SD = \sqrt{\text{Variance}}$

Calculation: $\sqrt{9} = 3$

Interpretation: On average, data points deviate from the mean by 3 units.

Question

Value = 85, Mean = 70, SD = 10. What is the z-score?

- A. 0.5
- B. 1.0
- C. 1.5
- D. 2.0

Answer: C

Explanation

Definition: The z-score (standard score) measures how many standard deviations a value is from the mean. It standardizes values for comparison across different distributions.

Formula: $z = (x - \mu) / \sigma$

Calculation: $(85 - 70) / 10 = 15 / 10 = 1.5$

Interpretation: The value 85 is 1.5 standard deviations above the mean.

Question

Value = 30, Min = 10, Max = 50. What is the min-max normalized value?

- A. 0.25
- B. 0.50
- C. 0.75
- D. 1.00

Answer: B

Explanation

Definition: Min-max normalization rescales values to a fixed range, typically [0, 1]. It preserves the relative relationships between data points.

Formula: $\text{Normalized} = (x - \text{min}) / (\text{max} - \text{min})$

Calculation: $(30 - 10) / (50 - 10) = 20 / 40 = 0.50$

Interpretation: The value 30 is exactly halfway between the minimum and maximum.

Question

Point A = (1, 2), Point B = (4, 6). What is the Euclidean distance?

- A. 3
- B. 4
- C. 5
- D. 7

Answer: C

Explanation

Definition: Euclidean distance is the straight-line distance between two points in Euclidean space. It is the most common distance metric in clustering algorithms like K-means.

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Calculation: $\sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Interpretation: The two points are 5 units apart in Euclidean space.

Question

Point A = (1, 2), Point B = (4, 6). What is the Manhattan distance?

- A. 3
- B. 5
- C. 7
- D. 9

Answer: C

Explanation

Definition: Manhattan distance (also called city-block or L1 distance) is the sum of the absolute differences between coordinates. It measures distance along grid lines, like walking in a city.

Formula: $d = |x_2 - x_1| + |y_2 - y_1|$

Calculation: $|4 - 1| + |6 - 2| = 3 + 4 = 7$

Interpretation: The Manhattan distance between the two points is 7 units.

Question

Points: (2, 4), (6, 8), (10, 12). What is the centroid?

- A. (4, 6)
- B. (6, 8)
- C. (8, 10)
- D. (9, 12)

Answer: B

Explanation

Definition: The centroid is the arithmetic mean of all points in a cluster. In K-means, the centroid represents the center of a cluster and is recalculated iteratively.

Formula: Centroid = (mean of x-coordinates, mean of y-coordinates)

Calculation: $((2+6+10)/3, (4+8+12)/3) = (18/3, 24/3) = (6, 8)$

Interpretation: The center of the cluster is at (6, 8).

Question

Points in cluster: (1, 1), (2, 2). Centroid = (1.5, 1.5). What is the WCSS?

- A. 0.5
- B. 1.0
- C. 2.0
- D. 4.0

Answer: B

Explanation

Definition: Within-Cluster Sum of Squares (WCSS) measures the compactness of a cluster. It is the sum of squared distances from each point to its cluster centroid. Lower WCSS indicates tighter, more compact clusters.

Formula: $WCSS = \sum d(\text{point}, \text{centroid})^2$

Calculation: $d_1^2 = (1-1.5)^2 + (1-1.5)^2 = 0.25+0.25 = 0.5$. $d_2^2 = (2-1.5)^2 + (2-1.5)^2 = 0.25+0.25 = 0.5$. $WCSS = 0.5+0.5 = 1.0$

Interpretation: The total squared deviation within the cluster is 1.0.

Question

$a = 2$, $b = 6$. What is the silhouette score?

- A. 0.33
- B. 0.50
- C. 0.67
- D. 0.75

Answer: C

Explanation

Definition: The silhouette score measures how similar a point is to its own cluster (cohesion, a) compared to the nearest other cluster (separation, b). It ranges from -1 to 1 . Higher values indicate better-defined clusters.

Formula: $s = (b - a) / \max(a, b)$

Calculation: $(6 - 2) / 6 = 4 / 6 = 0.67$

Interpretation: A score of 0.67 indicates reasonably good cluster separation.

Question

Point A = (0, 0), Point B = (3, 4). What is the Euclidean distance?

- A. 3
- B. 4
- C. 5
- D. 7

Answer: C

Explanation

Definition: Euclidean distance is the straight-line distance between two points.

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Calculation: $\sqrt{(3 - 0)^2 + (4 - 0)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Interpretation: The distance between the origin and (3, 4) is 5 units.

Question

Points: (0, 0), (4, 0), (2, 6). What is the centroid?

- A. (2, 2)
- B. (2, 3)
- C. (3, 2)
- D. (3, 3)

Answer: A

Explanation

Definition: The centroid is the arithmetic mean of all points in a cluster.

Formula: Centroid = (mean of x-coordinates, mean of y-coordinates)

Calculation: $((0+4+2)/3, (0+0+6)/3) = (6/3, 6/3) = (2, 2)$

Interpretation: The center of the cluster is at (2, 2).

Question

Point A = (2, 3), Point B = (5, 7). What is the Manhattan distance?

- A. 5
- B. 6
- C. 7
- D. 8

Answer: C

Explanation

Definition: Manhattan distance is the sum of absolute coordinate differences.

Formula: $d = |x_2 - x_1| + |y_2 - y_1|$

Calculation: $|5 - 2| + |7 - 3| = 3 + 4 = 7$

Interpretation: The Manhattan distance is 7 units.

Question

Points in cluster: (0, 0), (2, 0). Centroid = (1, 0). What is the WCSS?

- A. 1
- B. 2
- C. 4
- D. 8

Answer: B

Explanation

Definition: WCSS is the sum of squared distances from points to their centroid.

Formula: $WCSS = \sum d(\text{point}, \text{centroid})^2$

Calculation: $d_1^2 = (0-1)^2 + (0-0)^2 = 1$. $d_2^2 = (2-1)^2 + (0-0)^2 = 1$. $WCSS = 1+1 = 2$

Interpretation: The total squared deviation is 2.

Question

$a = 1$, $b = 5$. What is the silhouette score?

- A. 0.20
- B. 0.40
- C. 0.60
- D. 0.80

Answer: D

Explanation

Definition: The silhouette score measures cluster cohesion versus separation.

Formula: $s = (b - a) / \max(a, b)$

Calculation: $(5 - 1) / 5 = 4 / 5 = 0.80$

Interpretation: A score of 0.80 indicates strong cluster separation.

Question

Model: $y = 3 + 2x$. What is y when $x = 5$?

- A. 10
- B. 13
- C. 15
- D. 20

Answer: B

Explanation

Definition: In simple linear regression, the predicted value is the intercept plus the slope multiplied by the feature value.

Formula: $\hat{y} = \beta_0 + \beta_1 x$

Calculation: $y = 3 + 2(5) = 3 + 10 = 13$

Interpretation: When $x = 5$, the predicted y is 13.

Question

Model: $y = 10 - 1.5x$. What is y when $x = 4$?

- A. 2
- B. 4
- C. 6
- D. 8

Answer: B

Explanation

Definition: In simple linear regression, the predicted value is the intercept plus the slope multiplied by the feature value.

Formula: $\hat{y} = \beta_0 + \beta_1 x$

Calculation: $y = 10 - 1.5(4) = 10 - 6 = 4$

Interpretation: When $x = 4$, the predicted y is 4.

Question

Model: $y = 2 + 3x_1 - 1x_2$. What is y when $x_1 = 4$ and $x_2 = 5$?

- A. 7
- B. 9
- C. 11
- D. 13

Answer: B

Explanation

Definition: In multiple linear regression, the predicted value is the intercept plus the sum of each slope multiplied by its feature.

Formula: $\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2$

Calculation: $y = 2 + 3(4) - 1(5) = 2 + 12 - 5 = 9$

Interpretation: When $x_1 = 4$ and $x_2 = 5$, the predicted y is 9.

Question

Model: $z = -2 + 1x$. What is $P(y=1)$ when $x = 3$? (Use logistic function; $e \approx 2.718$)

- A. 0.27
- B. 0.50
- C. 0.73
- D. 0.88

Answer: C

Explanation

Definition: Logistic regression models the probability of a binary outcome using the logistic (sigmoid) function, which maps any real number to a value between 0 and 1.

Formula: $P(y=1) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1 x$

Calculation: $z = -2 + 1(3) = 1$. $P = 1/(1+e^{-1}) = 1/(1+0.368) = 1/1.368 = 0.73$

Interpretation: When $x = 3$, the probability of $y=1$ is 73 percent.

Question

Model: $z = 1 - 2x$. What is $P(y=1)$ when $x = 2$? (Use logistic function; $e \approx 2.718$)

- A. 0.05
- B. 0.12
- C. 0.27
- D. 0.50

Answer: A

Explanation

Definition: Logistic regression models the probability of a binary outcome using the logistic function.

Formula: $P(y=1) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1 x$

Calculation: $z = 1 - 2(2) = -3$. $P = 1/(1+e^3) = 1/(1+20.09) = 1/21.09 = 0.047 \approx 0.05$

Interpretation: When $x = 2$, the probability of $y=1$ is about 5 percent.

Question

TSS = 100, RSS = 20. What is R^2 ?

- A. 0.20
- B. 0.50
- C. 0.80
- D. 1.00

Answer: C

Explanation

Definition: R-squared (coefficient of determination) measures the proportion of variance in the dependent variable explained by the model. It ranges from 0 to 1, with higher values indicating better fit.

Formula: $R^2 = 1 - (RSS / TSS)$

Calculation: $1 - (20 / 100) = 1 - 0.20 = 0.80$

Interpretation: The model explains 80 percent of the variance in the target variable.

Question

TSS = 200, RSS = 50. What is R^2 ?

- A. 0.25
- B. 0.50
- C. 0.75
- D. 0.85

Answer: C

Explanation

Definition: R-squared measures the proportion of variance explained by the model.

Formula: $R^2 = 1 - (RSS / TSS)$

Calculation: $1 - (50 / 200) = 1 - 0.25 = 0.75$

Interpretation: The model explains 75 percent of the variance.

Question

$R^2 = 0.80$, $n = 50$, $k = 5$. What is adjusted R^2 ?

- A. 0.75
- B. 0.78
- C. 0.80
- D. 0.82

Answer: B

Explanation

Definition: Adjusted R-squared penalizes the addition of unnecessary predictors. It is always less than or equal to R-squared.

Formula: $\text{Adjusted } R^2 = 1 - [(1 - R^2)(n - 1) / (n - k - 1)]$

Calculation: $1 - [(1 - 0.80)(50 - 1) / (50 - 5 - 1)] = 1 - [0.20 \times 49 / 44] = 1 - 0.2227 = 0.7773 \approx 0.78$

Interpretation: After adjusting for the number of predictors, the model explains 78 percent of the variance.

Question

Actual = 25, Predicted = 22. What is the residual?

- A. -3
- B. 3
- C. 47
- D. 0.12

Answer: B

Explanation

Definition: A residual is the difference between the actual value and the predicted value. It represents the error of the model for a single observation.

Formula: $\text{Residual} = y - \hat{y}$

Calculation: $25 - 22 = 3$

Interpretation: The model under-predicted by 3 units.

Question

Actual = 18, Predicted = 20. What is the residual?

- A. -2
- B. 2
- C. 38
- D. 0.10

Answer: A

Explanation

Definition: A residual is the difference between actual and predicted values.

Formula: $\text{Residual} = y - \hat{y}$

Calculation: $18 - 20 = -2$

Interpretation: The model over-predicted by 2 units.